

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: DATA ANALYSIS WITH R PROGRAMMING

PRACTICAL 8

AIM: Performing two-way ANOVA using `aov()` (R).

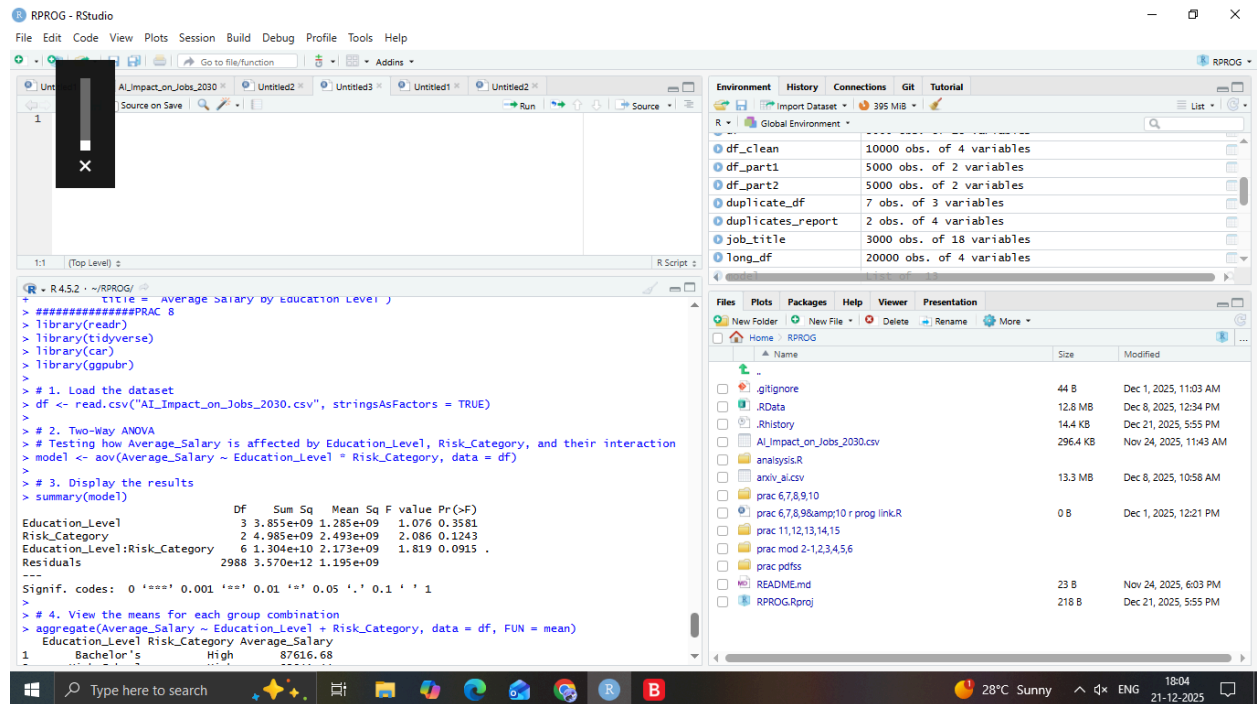
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The screenshot shows the RStudio interface with the following components:

- Environment:** Lists objects in the Global Environment, including `df_clean` (10000 obs. of 4 variables), `df_part1` (5000 obs. of 2 variables), `df_part2` (5000 obs. of 2 variables), `duplicate_df` (7 obs. of 3 variables), `duplicates_report` (2 obs. of 4 variables), `job_title` (3000 obs. of 18 variables), and `tong_df` (20000 obs. of 4 variables).
- Files:** Shows a file explorer with various files, including `AI_Impact_on_Jobs_2030.csv` (296.4 KB).
- Code Editor:** Contains the following R script:

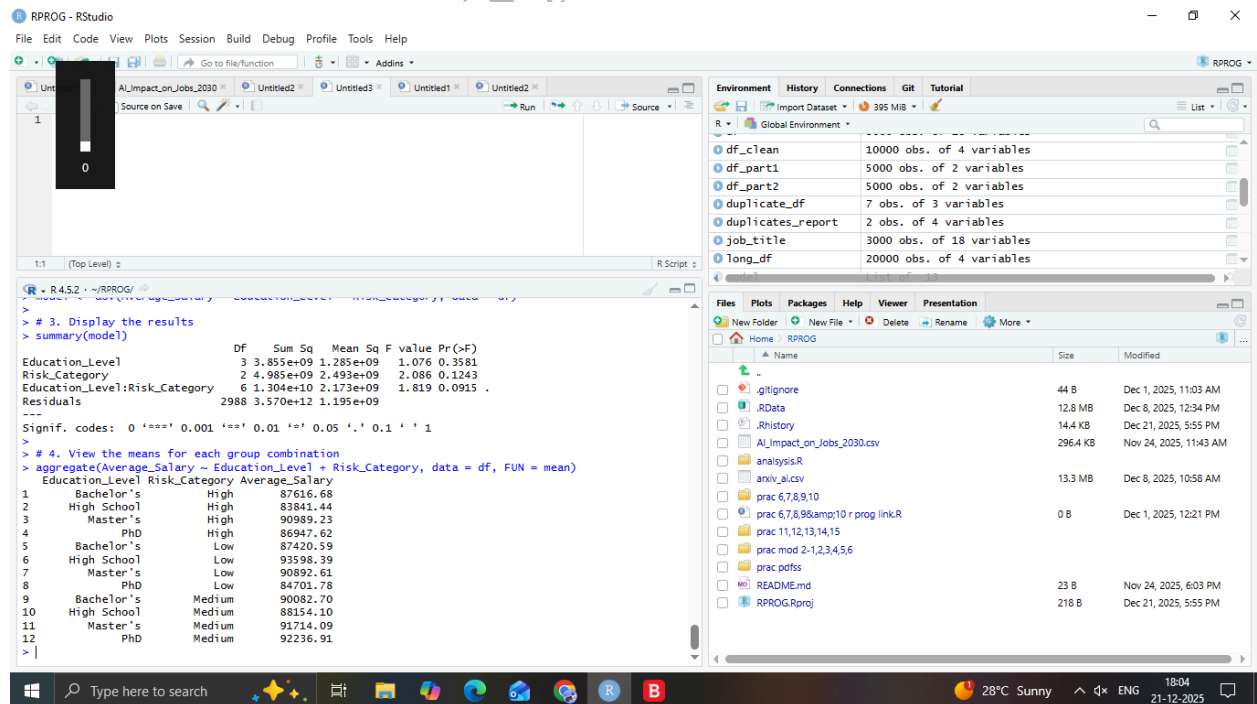
```
R> R 4.5.2 ~ ~/RPROG/
> title = Average Salary by Education Level )
> #####PRAC 8
> library(readr)
> library(tidyverse)
> library(car)
> library(ggpubr)
>
> # 1. Load the dataset
> df <- read_csv("AI_Impact_on_Jobs_2030.csv", stringsAsFactors = TRUE)
>
> # 2. Two-Way ANOVA
> # Testing how Average_Salary is affected by Education_Level, Risk_Category, and their interaction
> model <- aov(Average_Salary ~ Education_Level * Risk_Category, data = df)
>
> # 3. Display the results
> summary(model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Education_Level	3	3.855e+09	1.285e+09	1.076	0.3581
Risk_Category	2	4.985e+09	2.493e+09	2.086	0.1243
Education_Level:Risk_Category	6	1.304e+10	2.173e+09	1.819	0.0915
Residuals	2988	3.570e+12	1.195e+09		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> # 4. View the means for each group combination
> aggregate(Average_Salary ~ Education_Level * Risk_Category, data = df, FUN = mean)
```

Education_Level	Risk_Category	Average_Salary
1 Bachelor's	High	87616.68



The screenshot shows the RStudio interface with the following components:

- Environment:** Same as the first screenshot, listing objects in the Global Environment.
- Files:** Same as the first screenshot, showing a file explorer.
- Code Editor:** Contains the following R script:

```
R> R 4.5.2 ~ ~/RPROG/
> title = Average Salary by Education Level and Risk_Category )
> #####PRAC 8
> library(readr)
> library(tidyverse)
> library(car)
> library(ggpubr)
>
> # 1. Load the dataset
> df <- read_csv("AI_Impact_on_Jobs_2030.csv", stringsAsFactors = TRUE)
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```
> # 4. View the means for each group combination
> aggregate(Average_Salary ~ Education_Level * Risk_Category, data = df, FUN = mean)
```

Education_Level	Risk_Category	Average_Salary
1 Bachelor's	High	87616.68
2 High School	High	83841.44
3 Master's	High	90989.23
4 PHD	High	86947.62
5 Bachelor's	Low	87420.59
6 High School	Low	93598.39
7 Master's	Low	90892.61
8 PHD	Low	84701.78
9 Bachelor's	Medium	90082.70
10 High School	Medium	88154.10
11 Master's	Medium	91714.09
12 PHD	Medium	92236.91

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